



QUADRUPED ROBOT DEVELOPMENT

Certification-Based Industrial Training

WTL & MeitY Project | Group-E

Department of Electronics & Communication Engineering

Cooch Behar Government Engineering College | 2026

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Department of ECE, CGEC | WTL & MeitY Project | 2026

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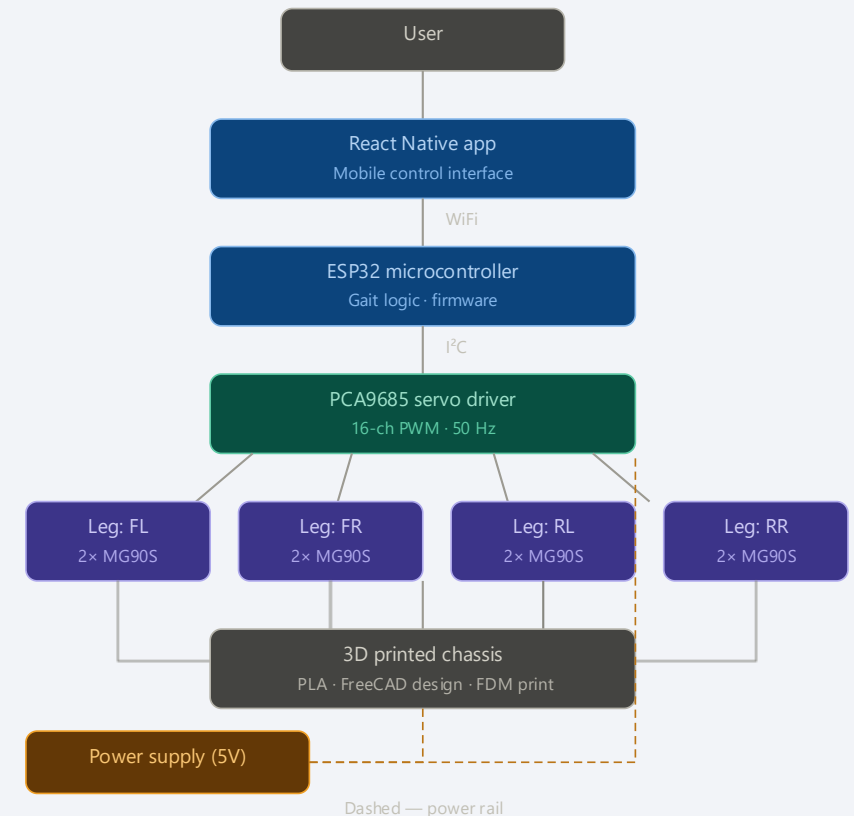
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Project Objective

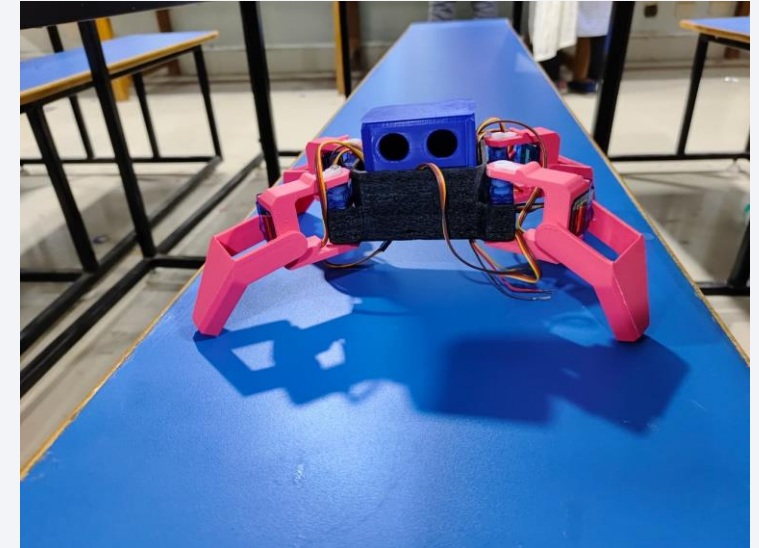
What are we building and why?

- ▶ Design, fabricate & program a fully functional 8-DOF Quadruped Robot
- ▶ Use affordable, open-source components and tools (ESP32 + FreeCAD + FDM printing)
- ▶ Core hardware: ESP32 microcontroller | PCA9685 16-ch servo driver | 8 × SG90S servos
- ▶ Chassis designed in FreeCAD — fabricated with FDM 3D printing (PLA+ & ABS)
- ▶ Hands-on experience across: Electronics • Mechanical Design • Additive Manufacturing
- ▶ End goal: Robot capable of standing stably and executing a walking gait
- ▶ Future roadmap: Sensor integration → Autonomous navigation
- ▶ Aligned with GoI skills development initiative under WTL & MeitY programme



Introduction — What is a Quadruped Robot?

- ▶ A legged mobile robot that moves on four limbs — mimicking dogs, horses, insects
- ▶ Unlike wheeled robots, quadrupeds navigate uneven terrain, climb stairs, step over obstacles
- ▶ Each leg has 2+ joints providing degrees of freedom for stable gait generation
- ▶ This robot: Hip joint + Knee joint per leg → 8 DOF for locomotion
- ▶ Notable examples: Boston Dynamics Spot • Unitree Go1 • University research platforms



8 DOF

Locomotion
(4 hips + 4 knees)

Static Head

Head
is stationary

8 Total

SG90S
Servos

3 Domains

Electronics
Mechanical
3D Printing

Real-World Applications of Quadruped Robots

Search & Rescue

Navigates rubble, collapsed structures and disaster zones where wheels cannot go

Rough Terrain

Pipeline inspection, mining surveys, and remote area monitoring

Military / Defence

Cargo carrying, reconnaissance missions, and hazardous environment scouting

Medical / Rehab

Companion and assistive robots for elderly care and rehabilitation

Research Platform

Studying locomotion algorithms, control theory, and AI-based navigation

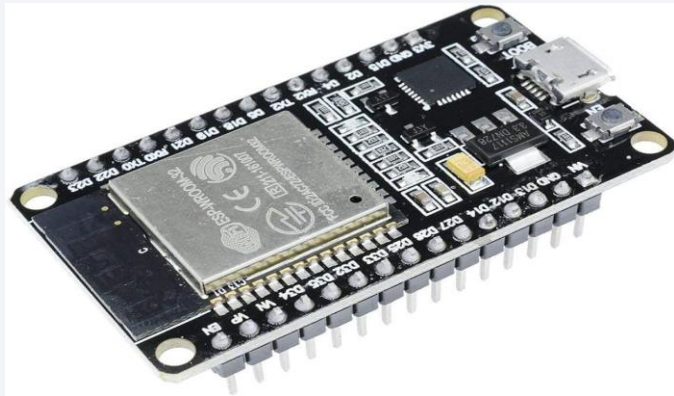
Industrial

Automated factory floor inspection and infrastructure monitoring

Why ESP32? & Why 3D Printing?

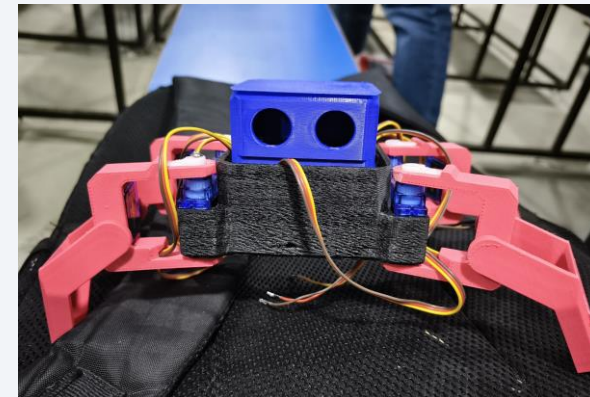
ESP32 — Central Controller

- ▶ Dual-core 240 MHz processor — powerful yet low-cost
- ▶ Built-in Wi-Fi & Bluetooth for wireless control
- ▶ Supports I2C and SPI for peripheral communication
- ▶ Well-supported Arduino IDE ecosystem
- ▶ ESP32Servo library — direct servo control via PWM
- ▶ Wireless capability → OTA updates, remote teleoperation



3D Printing — Chassis Fabrication

- ▶ Rapid iteration on part dimensions without tooling cost
- ▶ Customised mounting positions for every component
- ▶ Fabricate geometrically complex parts cheaply
- ▶ PLA+ → structural rigidity for Core Body & Head Box
- ▶ ABS → flexibility & impact resistance for leg links
- ▶ Entire robot chassis printed in the CGEC 3D Printing Lab



Development Roadmap — 9-Phase Journey

From component verification to autonomous navigation



 This report covers Phases 1–5 in detail. Phases 6–9 form the future roadmap.

Electronics Architecture

4-Layer electronics stack

Layer 1 — Central Controller

ESP32 Dev Board (I2C Master, Dual-core 240 MHz)

Layer 2 — Servo Driver

PCA9685 16-Channel PWM Driver (I2C Slave @ 0x40)

Layer 3 — Actuators

8× SG90S Metal Gear Micro Servos (4 Hip + 4 Knee)

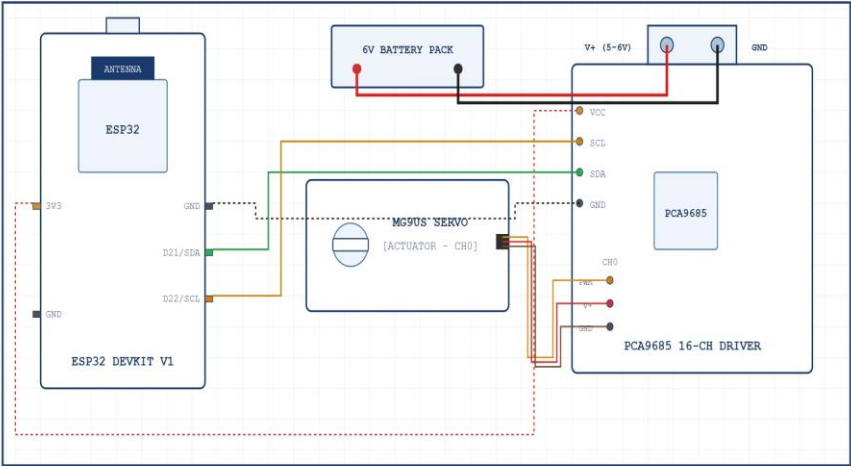
Layer 4 — Power Supply

External 5V Supply → PCA9685 V+ Servo Rail

⚠ PCA9685 V+ rail **MUST** have an external 5V supply (separate from VCC)

ESP32 → PCA9685 Wiring

ESP32 Pin	PCA9685 Pin
3.3V	VCC
GND	GND
GPIO21	SDA
GPIO22	SCL



Servo Channel Mapping

PCA9685 CH0–CH7 → 8 SG90S servos

CH0	Front Left Hip Type: Hip	CH5	Rear Left Knee Type: Knee
CH1	Front Left Knee Type: Knee	CH6	Rear Right Hip Type: Hip
CH2	Front Right Hip Type: Hip	CH7	Rear Right Knee Type: Knee
CH3	Front Right Knee Type: Knee		
CH4	Rear Left Hip Type: Hip		

Hip Servo

Knee Servo

Head Servo

Critical Discovery — Servo Power Rail Issue

Real-world debugging experience during Phase 1

SYMPTOM

All 8 servos were completely unresponsive when controlled via PCA9685

I2C DIAGNOSIS

I2C scanner confirmed PCA9685 at address 0x40 — ESP32, I2C bus, and chip all functioning correctly

ROOT CAUSE

V+ rail measured only 0.10 V (expected 4.8–5.2 V) — servo power rail NOT connected to external 5V supply

RESOLUTION

Connected 5V directly to PCA9685 V+ terminal → Servos responded immediately. V+ measured 4.92–5.05 V

LESSON

Always verify servo rail voltage FIRST before suspecting firmware or I2C communication issues

Mechanical Architecture & Component Dimensions

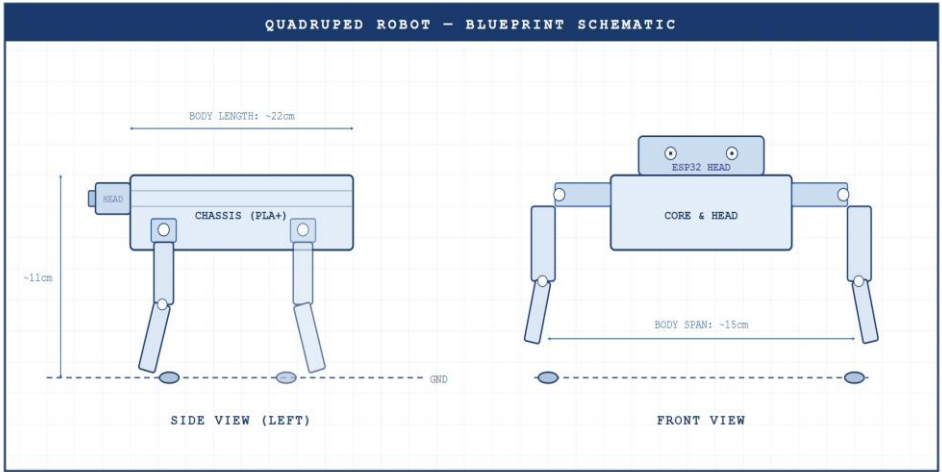
FreeCAD-designed chassis — all parts 3D printed

Structural Overview

- ▶ Two main assemblies: Head Box (electronics enclosure) + Core Body (central chassis)
- ▶ Head Box — houses ESP32, PCA9685, power switch + future sensor mount provision
- ▶ Core Body — mounts all 4 hip servos (4 total on chassis)
- ▶ Each leg: Upper Leg (hip-driven) + Lower Leg (knee-driven) → 2 links per leg

Component Dimensions

Component	Dimensions (mm)	Notes
Core Body	110 × 85 × 4	4 hip mounts
Head Box	85 × 65 × 45	Wall 2.5 mm — ESP32, PCA9685
Upper Leg ×4	45 × 10 × 5	PLA+ — hip servo driven
Lower Leg ×4	55 × 10 × 5	PLA+ — knee servo driven



FreeCAD Design Workflow

Part Design workbench — parametric modelling

1

Core Body

Base 110×85×4 mm plate with 4 symmetric hip servo pockets and central head servo mount

2

Head Box

Enclosure designed around ESP32 and PCA9685 PCB footprints (85×65×45 mm)

3

Hip Servo Mounts

Four symmetric press-fit pockets at FL, FR, RL, RR corners

4

Upper Leg

45 mm PLA link with servo horn connection hole at proximal end

5

Lower Leg

55 mm PLA link with foot pad boss at distal (terminal) end

6

Assembly

All bodies assembled in FreeCAD Assembly with correct spacing and constraints

4

3D Printing Process — FDM Technology

CAD → STL → G-code → Print

Fused Deposition Modelling (FDM): thermoplastic filament melted & deposited layer-by-layer | Workflow: CAD Model → STL Export → Slicer (G-code) → 3D Printer

Material Selection

ABS

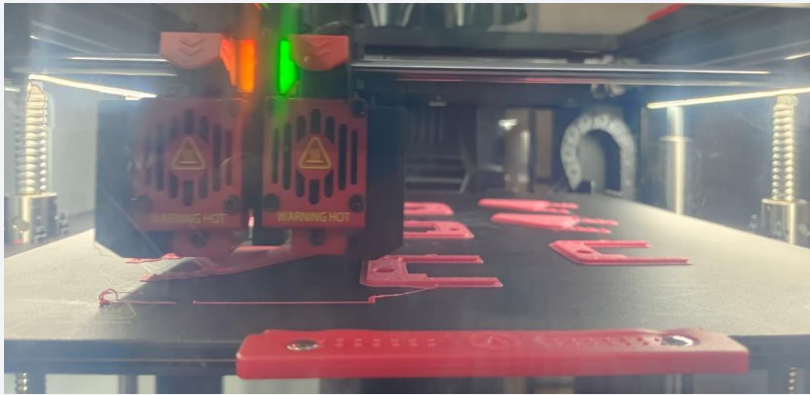
Core Body, Head Box

High stiffness, low warping, easy to print

PLA

Upper Leg, Lower Leg (×4 each)

Toughness & flexibility for repeated bending/impact loads

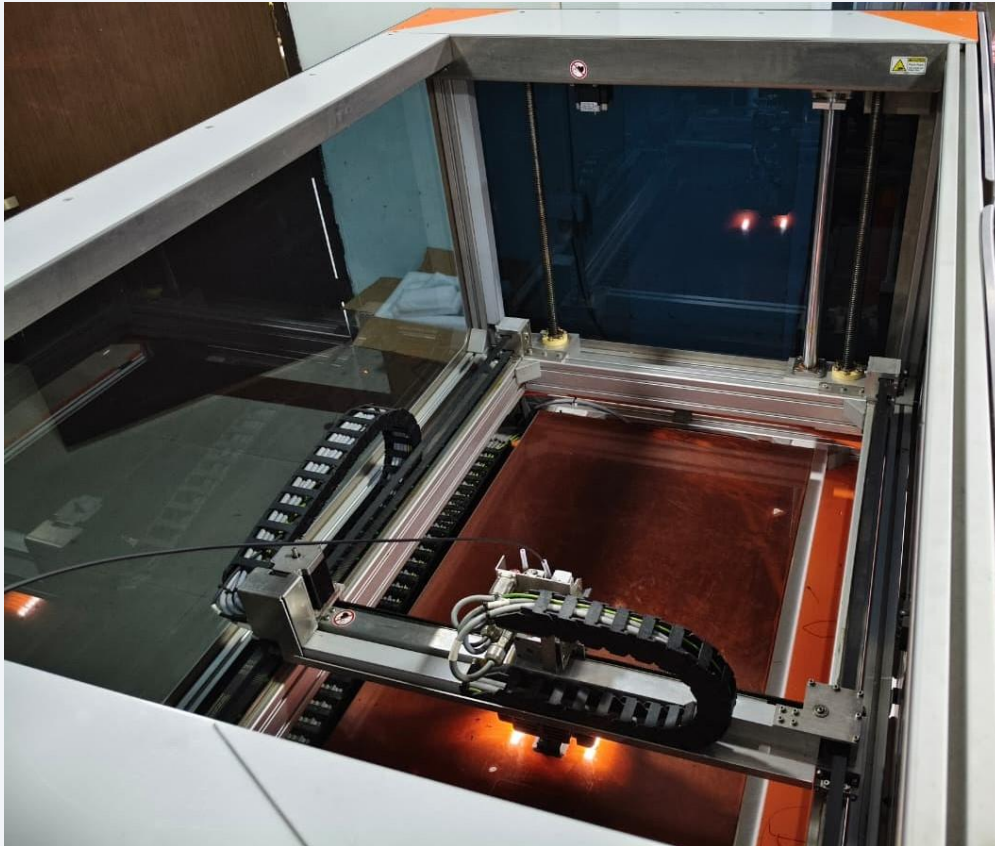


Slicer Settings

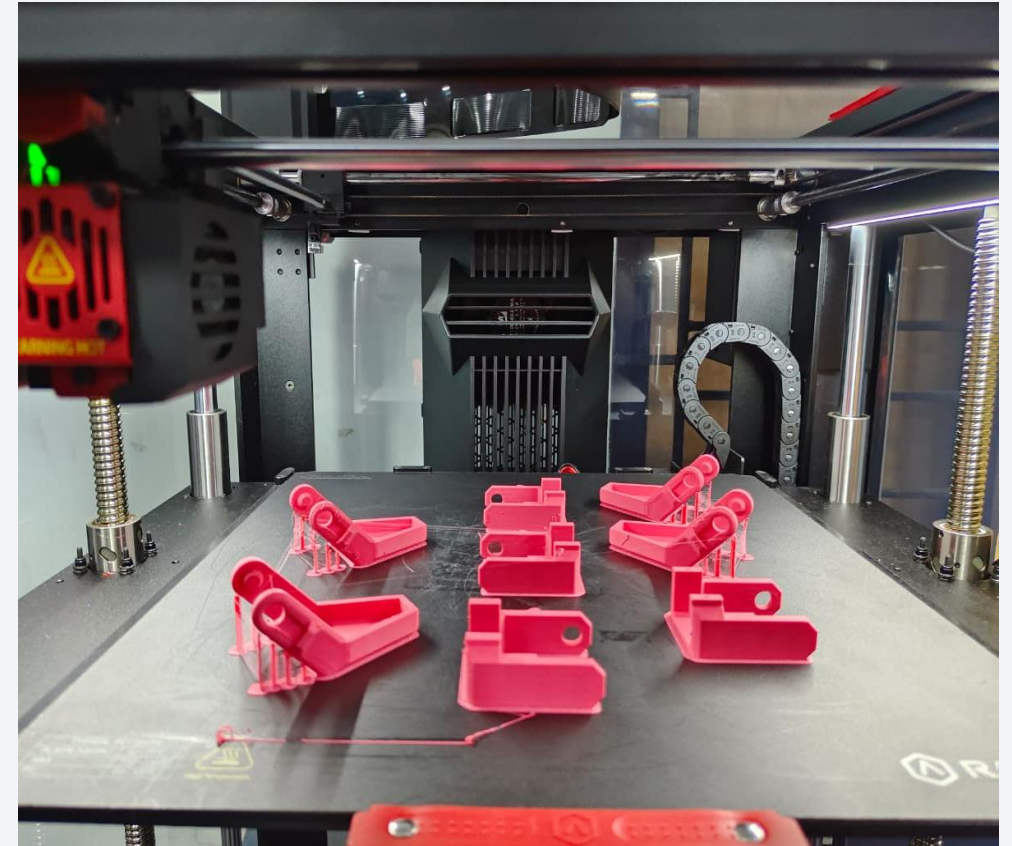
Parameter	Value
Layer Height	0.2 mm
Infill Density	40%
Perimeter Walls	4
Nozzle Diameter	0.4 mm
Print Speed	50 mm/s
Support	As required

3D Printing in the Lab

Fabrication of quadruped robot components at CGEC 3D Printing Lab



3D Printer (FDM) in the CGEC Lab during fabrication of the Core Body and Head Box using ABS.



Fabricated leg components (Upper and Lower leg segments) printed in magenta PLA for high layer adhesion and impact strength.

Products Made — 3D Printed Parts

19 total printed components

Part	Material	Dimensions	Qty	Description
Core Body	ABS	110×85×4 mm	1	Central chassis; 4 hip mounts; 4mm thickness for bending rigidity
Head Box	ABS	85×65×45 mm	1	Electronics enclosure for ESP32, PCA9685 & power switch; 2.5mm walls; future sensor provision
Upper Leg	PLA	45×10×5 mm	4	Hip-servo driven link; servo horn attachment hole at proximal end
Lower Leg	PLA	55×10×5 mm	4	Knee-servo driven link; foot pad boss at distal terminal end

Arduino Programming

Arduino IDE + ESP32 board support package

Development Environment & Libraries

ESP32Servo

Direct servo control via GPIO PWM

Wire

I2C communication for PCA9685

Adafruit_PWMServoDriver

High-level PCA9685 servo control

Code Listings

Listing 1

Servo Test Code: Direct GPIO18 sweep — 0° → 90° → 180° per servo

Listing 2

PCA9685 I2C Scanner: Scans I2C bus, confirms address 0x40

Listing 3

I2C Scan Output: Expected serial monitor: 'I2C device found at 0x40'

Listing 4

PCA9685 Servo Control: All-channel sweep through Adafruit PWM library

```
// Servo Test Code – ESP32 direct GPIO18
```

```
#include <ESP32Servo.h>
```

```
Servo myServo;
```

```
void setup() {
```

```
    myServo.attach(18); // GPIO18
```

```
}
```

```
void loop() {
```

```
    myServo.write(0);    delay(1000);
```

```
    myServo.write(90);   delay(1000);
```

```
    myServo.write(180);  delay(1000);
```

```
}
```


Results & Testing

I2C verification · Power rail · All-servo sweep

I2C Communication Verification

PCA9685 confirmed at address 0x40 via I2C scanner

Full chain validated: ESP32 → GPIO21/22 → SDA/SCL → PCA9685

Power Rail Diagnosis & Resolution

Before fix: V+ = 0.10 V (expected 4.8–5.2 V)

After fix: V+ = 4.92–5.05 V → All servos responded immediately

Individual Servo Test Results (All 8 Passed)

#	Function	0°	180°
1	Front Left Hip (CH0)	✓ Pass	✓ Pass
2	Front Left Knee (CH1)	✓ Pass	✓ Pass
3	Front Right Hip (CH2)	✓ Pass	✓ Pass
4	Front Right Knee (CH3)	✓ Pass	✓ Pass
5	Rear Left Hip (CH4)	✓ Pass	✓ Pass
6	Rear Left Knee (CH5)	✓ Pass	✓ Pass
7	Rear Right Hip (CH6)	✓ Pass	✓ Pass
8	Rear Right Knee (CH7)	✓ Pass	✓ Pass

Development Phase Status

Current progress across all 9 phases

1	Component Verification	Complete	6	Walking Gait	In Progress
2	CAD Design in FreeCAD	Complete	7	Turning Gait	Planned
3	3D Printing All Parts	Complete	8	Sensor Integration	Planned
4	Servo Calibration	Complete	9	Autonomous Navigation	Future
5	Standing Pose	Complete			



Conclusion

Key achievements and future roadmap

Electronics

Systematic debugging validated: I2C chain, servo power rail fix, all 8 servos operational across full 0°–180° range

Mechanical Design

Parametric FreeCAD CAD workflow mastered: , press-fit tolerances, STL export → design-for-manufacture skills

3D Printing

Hands-on material selection (PLA & ABS), slicer optimization, and print quality assessment in CGEC Lab

Overall Platform

Fully functional 8-DOF quadruped hardware + software stack built — validated foundation for walking gait development

Roadmap: Phase 6 Walking Gait (IK) → Phase 7 Turning → Phase 8 Sensors → Phase 9 Autonomous Navigation via ESP32 Wi-Fi

References

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- [3] NXP Semiconductors, PCA9685 16-channel 12-bit PWM I2C-bus LED controller, Product datasheet Rev. 4, NXP, 2015
- [4] Tower Pro, SG90S Metal Gear Micro Servo Datasheet, Tower Pro Co. Ltd., 2020
- [5] FreeCAD Community, FreeCAD: Your Own 3D Parametric Modeler, <https://www.freecad.org>

THANK YOU

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WTL & MeitY Industrial Training Programme

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